

# ImprovEvolve on the Cohn Catalogue: a Calibrated General Spherical-Code Improver

## Summary

We evolved a general spherical-code improver (dimensions  $d \in [8, 16]$ ) and benchmark it against the Cohn catalogue and the two prior paper programs (E7, E8). The headline metric is the mean relative improvement of the minimal achievable maximum pairwise inner product  $\mu = \max_{i < j} \langle x_i, x_j \rangle$  over Cohn, floored at 0 per configuration:  $\% = 100 \cdot \text{mean}_{\text{configs}} \max(0, (\mu_{\text{Cohn}} - \mu_{\text{best}})/|\mu_{\text{Cohn}}|)$ . Parameters were tuned on a 14-configuration validation *panel* (the high-headroom configs that steered evolution) and then applied unchanged to the full 90-configuration *test* set, identically for every program (a fair head-to-head).

**Headline (full90 test, protocol  $P^*$ , per-config wall 3540s, best of 3 seeds).** Our champion reaches **0.1765%** mean relative improvement over Cohn (53/90 configs improved, 90/90 valid), **2.35 $\times$  the best prior program** (E7 0.0752%, E8 0.0618%). The advantage concentrates in the high-headroom dimensions  $d = 13$  and  $d = 16$ .

## Calibration (validation panel)

**Metric replication.** Under the canonical grader (`validate.py`; single monotone  $L$ -BFGS walk,  $R=1$ ,  $B=10$ , seed 42) at the evolution’s true per-config budget (3540s) the champion scores 0.4645% on the panel — reproducing the plotted  $\sim 0.46$ . Fitness is wall-limited and monotone in compute (0.4125% at 600s  $\rightarrow$  0.4645% at 3540s).

**Protocol sweep.** Sweeping the basin-hopping knobs on the panel (noising steps  $M$ , noise schedule  $\sigma_{\text{max}} \rightarrow \sigma_{\text{min}}$ , fresh-restart period, restarts) selects  $P^*$ : *unbounded restarts*,  $M=10$ ,  $\sigma: 1 \rightarrow 10^{-6}$ , *fresh every 5*. Fewer noising steps per restart let more restarts fit a fixed wall, yielding more basin hops; the fresh-restart period is irrelevant; high base-seed variance at short walls collapses as the wall grows.

**Beating canonical at equal budget.** At an identical 3540s per-config wall on the panel,  $P^*$  (restart-heavy) scores **0.4797%** vs the canonical single-chain 0.4645% — a +3.3% relative gain purely from the search protocol.

## Per-dimension improvement (full90, % over Cohn)

| $d$ | E7     | E8     | ours          |
|-----|--------|--------|---------------|
| 8   | 0.0446 | 0.0403 | <b>0.0645</b> |
| 9   | 0.0887 | 0.1218 | <b>0.1723</b> |
| 10  | 0.0890 | 0.0205 | <b>0.1097</b> |
| 11  | 0.0832 | 0.0667 | <b>0.1191</b> |
| 12  | 0.0560 | 0.0533 | <b>0.2239</b> |
| 13  | 0.1562 | 0.1605 | <b>0.4190</b> |
| 14  | 0.0057 | 0.0002 | <b>0.0428</b> |
| 15  | 0.1138 | 0.0473 | <b>0.1241</b> |
| 16  | 0.0401 | 0.0459 | <b>0.3133</b> |
| all | 0.0752 | 0.0618 | <b>0.1765</b> |

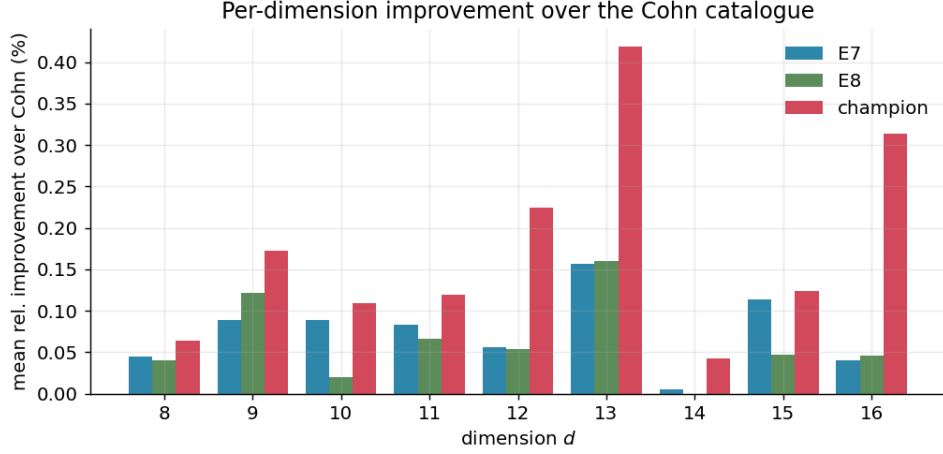


Figure 1: Per-dimension mean relative improvement over the Cohn catalogue for E7, E8 and our champion (full90 test).

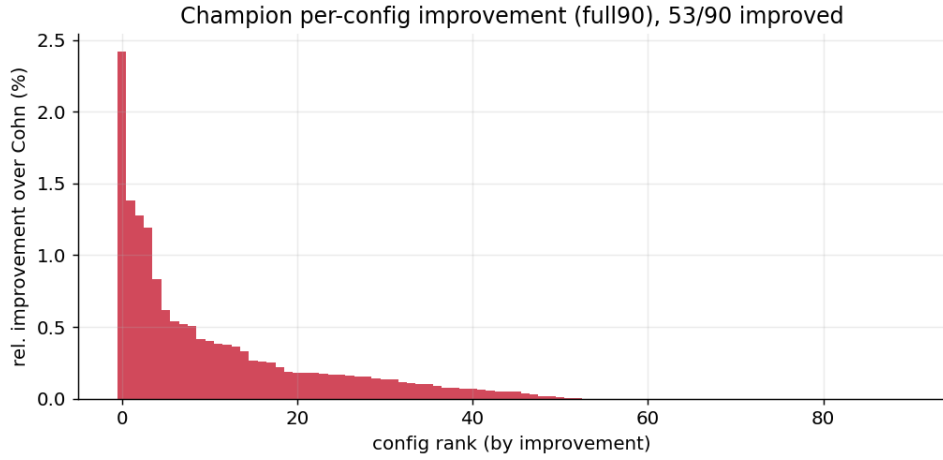


Figure 2: Per-configuration improvement of the champion over Cohn, sorted; bars above zero are configs where we strictly beat the catalogue.

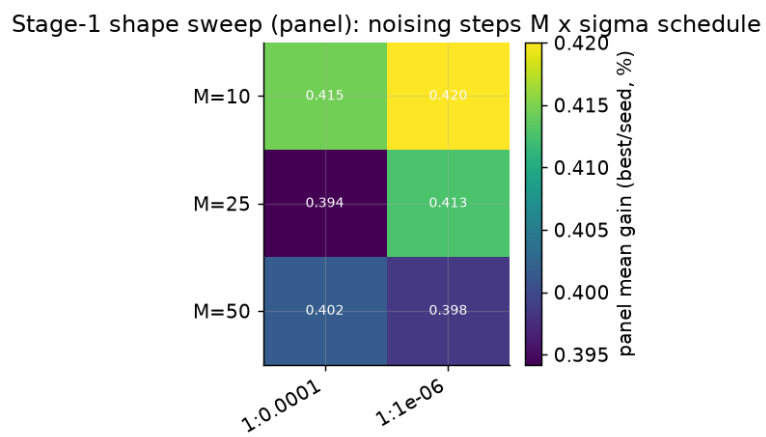


Figure 3: Validation-panel calibration: best-over-seed mean gain as a function of noising steps  $M$  and noise schedule. Smaller  $M$  (more restarts per wall) wins.

## Qualitative structure (extreme-gain configurations)

UMAP projection of unit vectors: Cohn vs ImprovEvolve champion (largest-improvement configurations)

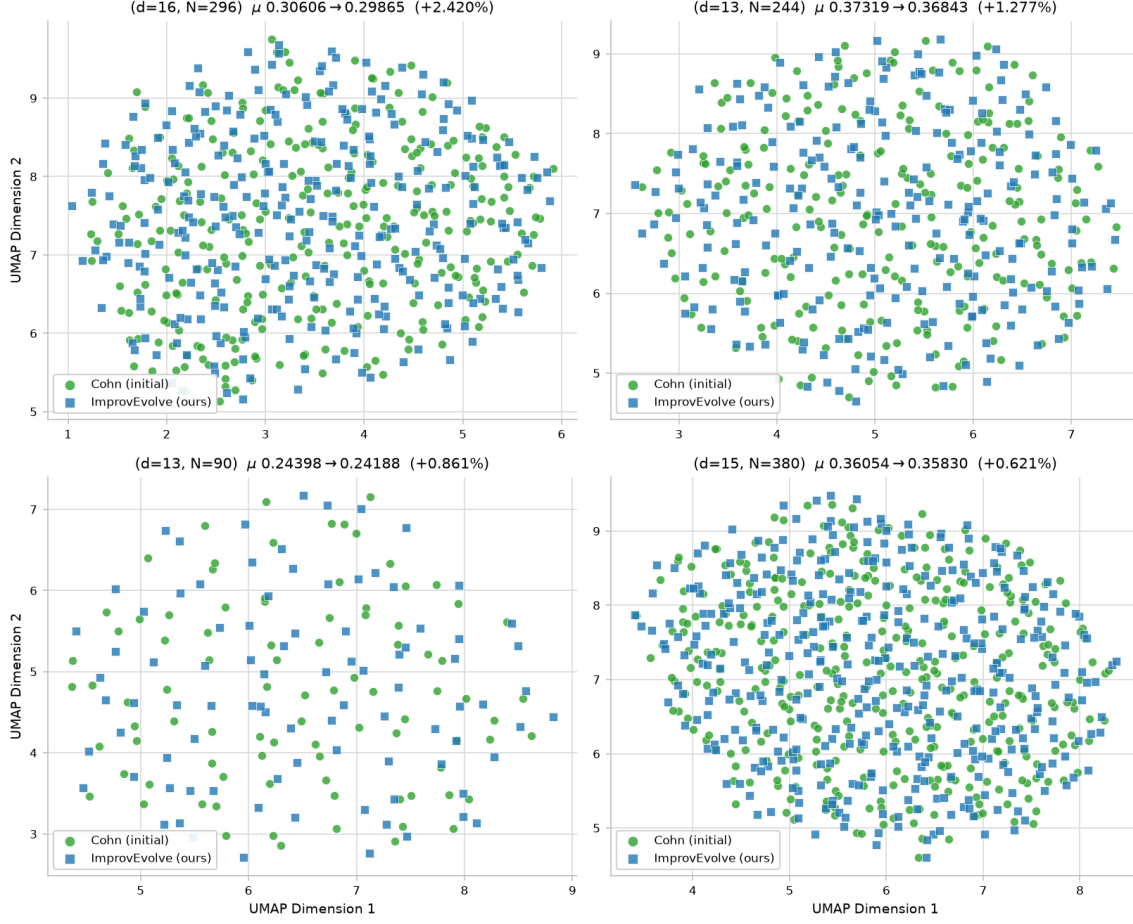


Figure 4: UMAP (cosine) of the unit vectors for the four configurations with the largest champion improvement: Cohn (green circles) vs. ImprovEvolve champion (blue squares), a single shared embedding per panel. The two distributions overlap because both are near-uniform on the same sphere  $S^{d-1}$ : every point is in fact relocated (mean displacement  $\approx 0.3$ – $0.6$ , up to near-antipodal), but the improvement lives in the fine pairwise-extreme arrangement rather than the marginal density, so the catalogue and the improved code occupy the same embedded region.

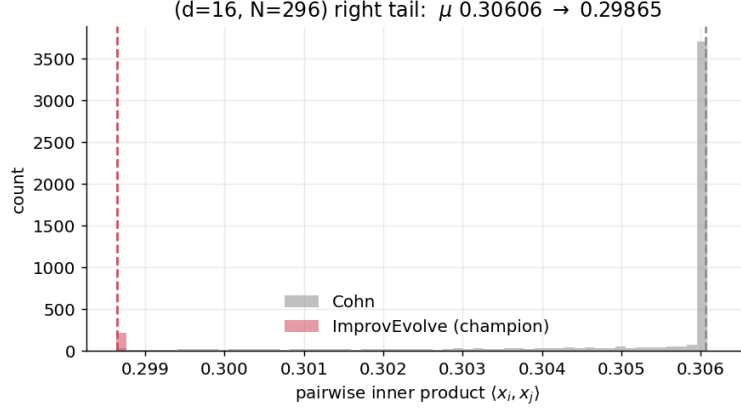


Figure 5: Pairwise inner-product right tail, Cohn vs champion, ( $d=16, N=296$ ). Dashed lines mark  $\mu = \max_{i < j} \langle x_i, x_j \rangle$ .

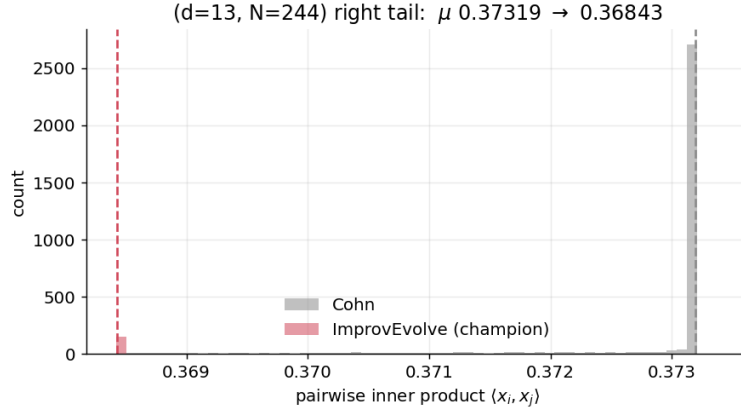


Figure 6: Pairwise inner-product right tail, ( $d=13, N=244$ ).

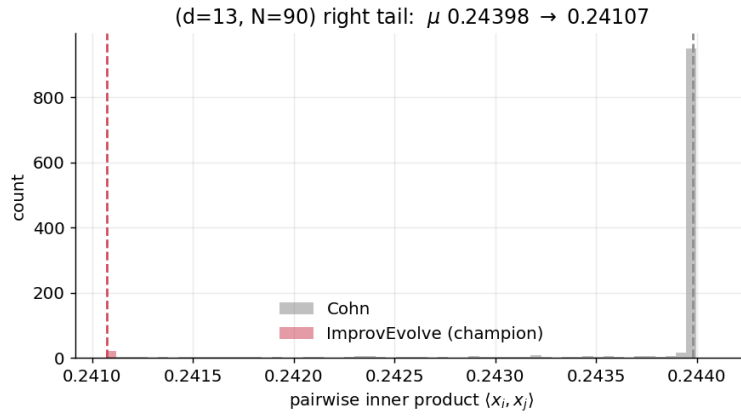


Figure 7: Pairwise inner-product right tail, ( $d=13, N=90$ ).

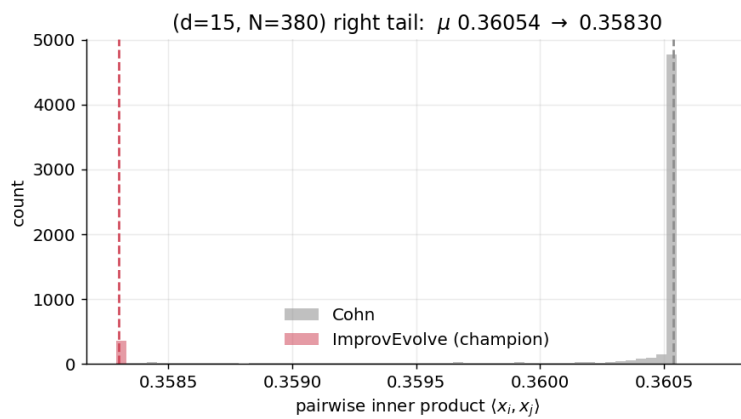


Figure 8: Pairwise inner-product right tail, ( $d=15, N=380$ ).

## Full 90-configuration table

Table 1: Full 90-configuration head-to-head: minimal achievable maximum pairwise inner product  $\mu = \max_{i < j} \langle x_i, x_j \rangle$  (lower is better) under the calibrated protocol  $P^*$ . **Bold** = strictly beats the Cohn catalogue. Summary row: mean relative improvement over Cohn (# configs improved).

| $d$ | $N$ | $\mu_{\text{Cohn}}$ | $\mu_{\text{E7}}$ | $\mu_{\text{E8}}$ | $\mu_{\text{ours}}$ |
|-----|-----|---------------------|-------------------|-------------------|---------------------|
| 8   | 26  | 0.172652            | 0.172652          | 0.172652          | 0.172652            |
| 8   | 51  | 0.316443            | 0.316443          | 0.316443          | 0.316443            |
| 8   | 86  | 0.405335            | 0.405335          | 0.405335          | 0.405335            |
| 8   | 283 | 0.582807            | 0.582807          | 0.582807          | 0.582 <b>428</b>    |
| 8   | 322 | 0.597017            | 0.597017          | 0.597017          | 0.597017            |
| 8   | 528 | 0.655771            | 0.65 <b>4787</b>  | 0.655 <b>037</b>  | 0.65 <b>4905</b>    |
| 8   | 656 | 0.667356            | 0.667 <b>005</b>  | 0.667356          | 0.666 <b>608</b>    |
| 8   | 669 | 0.670139            | <b>0.669282</b>   | <b>0.668808</b>   | <b>0.669239</b>     |
| 8   | 835 | 0.699340            | <b>0.698933</b>   | 0.699340          | 0.699 <b>200</b>    |
| 8   | 873 | 0.703836            | 0.703 <b>430</b>  | 0.703 <b>182</b>  | 0.702 <b>561</b>    |
| 9   | 32  | 0.150221            | 0.150221          | 0.150221          | 0.150221            |
| 9   | 44  | 0.233555            | 0.233555          | 0.233555          | 0.233555            |
| 9   | 521 | 0.594856            | 0.59 <b>2695</b>  | 0.59 <b>2584</b>  | 0.589 <b>902</b>    |
| 9   | 625 | 0.616874            | 0.61 <b>4612</b>  | 0.61 <b>4603</b>  | 0.613 <b>733</b>    |
| 9   | 750 | 0.646038            | 0.645 <b>774</b>  | 0.645 <b>644</b>  | 0.645 <b>669</b>    |
| 9   | 786 | 0.650670            | 0.650 <b>441</b>  | 0.649 <b>881</b>  | 0.650 <b>086</b>    |
| 9   | 838 | 0.656242            | 0.656 <b>120</b>  | 0.655 <b>753</b>  | 0.655 <b>902</b>    |
| 9   | 869 | 0.659627            | 0.659 <b>420</b>  | 0.659 <b>090</b>  | 0.659 <b>103</b>    |
| 9   | 880 | 0.660630            | 0.660 <b>528</b>  | 0.660 <b>216</b>  | 0.660 <b>293</b>    |
| 9   | 886 | 0.661239            | 0.661 <b>137</b>  | 0.660 <b>794</b>  | 0.660 <b>899</b>    |
| 10  | 91  | 0.326529            | 0.326 <b>351</b>  | 0.326 <b>295</b>  | 0.326 <b>304</b>    |
| 10  | 136 | 0.393476            | 0.3934 <b>29</b>  | 0.3933 <b>47</b>  | 0.392 <b>930</b>    |
| 10  | 271 | 0.480324            | 0.480 <b>160</b>  | 0.480 <b>277</b>  | 0.478 <b>526</b>    |
| 10  | 314 | 0.492047            | 0.491 <b>805</b>  | 0.492047          | 0.491 <b>275</b>    |
| 10  | 324 | 0.494738            | 0.494 <b>438</b>  | 0.494 <b>289</b>  | 0.493 <b>816</b>    |
| 10  | 785 | 0.575536            | 0.573 <b>796</b>  | 0.575536          | 0.574 <b>547</b>    |
| 10  | 827 | 0.588319            | 0.587 <b>660</b>  | 0.588319          | 0.588319            |
| 10  | 862 | 0.593249            | 0.59 <b>2531</b>  | 0.593249          | 0.593249            |
| 10  | 868 | 0.594110            | 0.593 <b>250</b>  | 0.594110          | 0.594110            |
| 10  | 912 | 0.598288            | 0.598288          | 0.598288          | 0.598288            |
| 11  | 84  | 0.265219            | 0.265 <b>035</b>  | 0.265 <b>197</b>  | 0.265 <b>033</b>    |
| 11  | 268 | 0.439263            | 0.438 <b>967</b>  | 0.439263          | 0.438 <b>515</b>    |
| 11  | 352 | 0.477385            | 0.477385          | 0.477385          | 0.477385            |
| 11  | 373 | 0.482230            | 0.481 <b>803</b>  | 0.482 <b>130</b>  | 0.480 <b>479</b>    |
| 11  | 407 | 0.487150            | 0.486 <b>685</b>  | 0.487150          | 0.486 <b>263</b>    |
| 11  | 521 | 0.499653            | 0.499653          | 0.499653          | 0.499653            |
| 11  | 590 | 0.499977            | 0.499 <b>711</b>  | 0.499 <b>694</b>  | 0.499 <b>886</b>    |
| 11  | 613 | 0.513245            | 0.51 <b>2363</b>  | 0.513245          | 0.513245            |
| 11  | 692 | 0.542819            | 0.541 <b>982</b>  | 0.541 <b>851</b>  | 0.541 <b>923</b>    |
| 11  | 740 | 0.554281            | 0.553 <b>552</b>  | 0.552 <b>044</b>  | 0.553 <b>051</b>    |
| 12  | 95  | 0.265091            | 0.265091          | 0.265091          | 0.265091            |
| 12  | 240 | 0.398274            | 0.397 <b>952</b>  | 0.398274          | 0.392 <b>776</b>    |
| 12  | 331 | 0.430981            | 0.430981          | 0.430981          | 0.430981            |
| 12  | 343 | 0.432985            | 0.431 <b>217</b>  | 0.431 <b>049</b>  | 0.430 <b>647</b>    |
| 12  | 416 | 0.451180            | 0.451 <b>096</b>  | 0.451180          | 0.450 <b>365</b>    |
| 12  | 642 | 0.493028            | 0.492 <b>858</b>  | 0.492 <b>802</b>  | 0.492 <b>537</b>    |
| 12  | 734 | 0.496809            | 0.496 <b>723</b>  | 0.496 <b>609</b>  | 0.496 <b>617</b>    |
| 12  | 855 | 0.520373            | 0.520373          | 0.520373          | 0.520373            |
| 12  | 899 | 0.532521            | 0.532521          | 0.532521          | 0.532521            |
| 12  | 913 | 0.535266            | 0.535266          | 0.535266          | 0.535266            |
| 13  | 69  | 0.208427            | 0.208418          | 0.207 <b>659</b>  | 0.207 <b>594</b>    |
| 13  | 90  | 0.243979            | 0.243904          | 0.243979          | 0.241 <b>074</b>    |
| 13  | 162 | 0.313684            | 0.313684          | 0.3136 <b>73</b>  | 0.312 <b>475</b>    |
| 13  | 244 | 0.373192            | 0.373 <b>015</b>  | 0.371 <b>484</b>  | 0.368 <b>425</b>    |
| 13  | 351 | 0.412084            | 0.411 <b>518</b>  | 0.411 <b>583</b>  | 0.411 <b>023</b>    |
| 13  | 592 | 0.465094            | 0.461 <b>594</b>  | 0.465094          | 0.465094            |
| 13  | 691 | 0.482881            | 0.482 <b>536</b>  | 0.482 <b>555</b>  | 0.482 <b>521</b>    |
| 13  | 725 | 0.485536            | 0.484 <b>988</b>  | 0.485181          | 0.484 <b>966</b>    |

| $d$                                       | $N$  | $\mu_{\text{Cohn}}$ | $\mu_{\text{E7}}$ | $\mu_{\text{E8}}$ | $\mu_{\text{ours}}$ |
|-------------------------------------------|------|---------------------|-------------------|-------------------|---------------------|
| 13                                        | 818  | 0.490444            | <b>0.489433</b>   | <b>0.489567</b>   | <b>0.489671</b>     |
| 13                                        | 882  | 0.497360            | <b>0.496367</b>   | <b>0.495696</b>   | <b>0.495720</b>     |
| 14                                        | 102  | 0.228290            | 0.228290          | 0.228290          | <b>0.227337</b>     |
| 14                                        | 211  | 0.315963            | 0.315963          | 0.315963          | 0.315963            |
| 14                                        | 212  | 0.316396            | 0.316396          | <b>0.316390</b>   | <b>0.316367</b>     |
| 14                                        | 642  | 0.434187            | 0.434187          | 0.434187          | <b>0.434181</b>     |
| 14                                        | 692  | 0.454025            | <b>0.453768</b>   | 0.454025          | 0.454025            |
| 14                                        | 913  | 0.455197            | 0.455197          | 0.455197          | 0.455197            |
| 14                                        | 922  | 0.455345            | 0.455345          | 0.455345          | 0.455345            |
| 14                                        | 925  | 0.455352            | 0.455352          | 0.455352          | 0.455352            |
| 14                                        | 970  | 0.455616            | 0.455616          | 0.455616          | 0.455616            |
| 14                                        | 1021 | 0.457575            | 0.457575          | 0.457575          | 0.457575            |
| 15                                        | 208  | 0.265531            | 0.265531          | 0.265531          | 0.265531            |
| 15                                        | 243  | 0.286369            | 0.286369          | 0.286369          | 0.286369            |
| 15                                        | 380  | 0.360536            | <b>0.358658</b>   | <b>0.358831</b>   | <b>0.358298</b>     |
| 15                                        | 425  | 0.371377            | <b>0.371353</b>   | 0.371377          | <b>0.370433</b>     |
| 15                                        | 436  | 0.376463            | 0.376463          | 0.376463          | 0.376463            |
| 15                                        | 456  | 0.376463            | 0.376463          | 0.376463          | 0.376463            |
| 15                                        | 491  | 0.393061            | <b>0.392599</b>   | 0.393061          | 0.393061            |
| 15                                        | 514  | 0.395538            | <b>0.394835</b>   | 0.395538          | <b>0.395142</b>     |
| 15                                        | 517  | 0.396049            | <b>0.395548</b>   | 0.396049          | 0.396049            |
| 15                                        | 527  | 0.397630            | <b>0.396878</b>   | 0.397630          | <b>0.396573</b>     |
| 16                                        | 88   | 0.168668            | 0.168668          | 0.168668          | 0.168668            |
| 16                                        | 126  | 0.199994            | 0.199994          | 0.199994          | 0.199994            |
| 16                                        | 160  | 0.204975            | 0.204975          | 0.204975          | 0.204975            |
| 16                                        | 197  | 0.249964            | 0.249964          | 0.249964          | <b>0.249894</b>     |
| 16                                        | 296  | 0.306058            | <b>0.305908</b>   | <b>0.304967</b>   | <b>0.298652</b>     |
| 16                                        | 341  | 0.330381            | <b>0.330366</b>   | <b>0.330079</b>   | <b>0.329853</b>     |
| 16                                        | 534  | 0.367437            | 0.367437          | <b>0.367396</b>   | <b>0.367414</b>     |
| 16                                        | 688  | 0.397260            | 0.397260          | 0.397260          | 0.397260            |
| 16                                        | 770  | 0.400000            | 0.400000          | 0.400000          | 0.400000            |
| 16                                        | 807  | 0.411522            | <b>0.410094</b>   | 0.411522          | <b>0.409386</b>     |
| <i>mean rel. improvement (# improved)</i> |      | 0.0752% (54/90)     | 0.0618% (36/90)   | <b>0.1765%</b>    | <b>(53/90)</b>      |